

CSCE 671 HUMAN-COMPUTER INTERACTION · MODULE 02 · LECTURE NOTES

Perception

What people see, which is not what is on the screen.

MODULE OUTCOMES

Explain the grouping principles and apply them to a layout.

Explain contrast, colour, and their limits.

Explain preattentive features and pop-out.

Explain change blindness and its design consequences.

Predict what will and will not be noticed.

1 Grouping

PROXIMITY things close together are one group.
The strongest of these by a wide margin,
and the cheapest to use.

COMMON REGION a shared enclosure or background
groups, and it overrides proximity.

SIMILARITY same colour, shape, or size groups.
Weaker than proximity, and useful
across distance where proximity cannot
reach.

CONNECTEDNESS a visible link groups very strongly,
and overrides similarity.

CONTINUITY aligned things read as a sequence,
which is why alignment matters more
than it appears to.

CLOSURE the eye completes an implied shape, so
a partial border still encloses.

AND THE PRACTICAL CONSEQUENCE: if your spacing
contradicts your intended grouping, the spacing wins.
A label closer to the wrong field is attached to the
wrong field, whatever the markup says.

If the spacing contradicts the intended grouping, the spacing wins — which makes whitespace a functional rather than an aesthetic decision, and is the module's most immediately usable finding.

The ordering by strength matters too: when two principles conflict, the stronger one determines what the user perceives, which is why a shared background can hold together items that are not particularly close.

WHICH MAKES WHITESPACE STRUCTURAL RATHER THAN DECORATIVE

Grouping by proximity is automatic and pre-conscious — the user does not decide to group things, and cannot decide not to, which means the layout is communicating whether or not you intended it to.

So the layout communicates a structure whether or not you designed one, and the commonest form-design defect is a label placed equidistant between two fields, which attaches to neither reliably and produces errors that look like carelessness.

And the fix is free: reduce the space between related items and increase it between groups — which costs nothing, requires no new elements, adds no visual weight, and is the highest-return edit available in most layouts.

Which is why a cluttered interface is not merely ugly but genuinely harder to use — uniform spacing conveys no structure at all, so the user has to construct the structure from the content, which costs attention they could have spent on the task (Module 03 §1).

2 Contrast and colour

- **Colour is excellent for categorical distinction — up to about five to seven categories, after which people cannot hold the mapping** (Module 03 §2's working memory limit) and have to consult the legend for every item, which defeats the purpose.
- **And it is poor for ordered quantities** unless the scale is specifically designed for perceptual uniformity — **which is CSCE 679 Module 04's whole subject**, and where the default rainbow scale does real damage.
- **Luminance contrast carries the structure, and hue does not** — the visual system's spatial acuity is largely driven by luminance — **which is why a design must work in greyscale** before any colour is added to it.
- **And a substantial minority of people cannot distinguish certain hue pairs**, red and green most commonly — so **colour must never be the only channel carrying information** (CSCE 632 Module 03's requirement, arriving here as a perceptual fact rather than a compliance one).
- **Which makes redundant encoding the rule: colour and shape, colour and position, colour and a text label. 'Must work in greyscale' is the quickest available test, and it catches the colour-only encoding, the insufficient contrast, and the hue-carrying-the-structure error all at once** — which makes it worth running on every screen.

3 Preattentive features

A FEW VISUAL FEATURES ARE PROCESSED IN PARALLEL, SO ONE ITEM DIFFERING IN THEM IS FOUND INSTANTLY

Colour, size, orientation, motion, curvature, and a handful of others — a single item differing in one of these is located in approximately constant time, regardless of how many distractors are present, which is a striking and well-replicated result.

Which is the basis of every effective visual highlight in an interface or a chart — and critically, it only works for one feature at a time.

Because a conjunction of features requires serial search: finding the one red circle among red squares and blue circles takes time proportional to the number of items, because no single channel distinguishes the target — which is a measured and reliable result and is the reason the one-dimension rule holds.

So the design rule is: highlight along exactly one dimension — and the more things you highlight, the less any of them pops out, which is why an interface where everything is emphasised has no emphasis at all and is a countable rather than a stylistic defect.

- **Pick one feature for your single most important distinction, and do not reuse that feature for anything else** — which is a discipline about the whole screen rather than about one element.
- **Motion is the strongest channel and the most expensive — it captures attention involuntarily**, which is simultaneously **why an animated advertisement works and why it is intolerable**, and is why it should be reserved for genuinely urgent things.
- **And count your emphasis. If more than about three things on a screen are emphasised, none of them is** — which makes this a **countable defect** you can check in a design review without any argument about taste.
- **Reserve the strongest available channel for the rarest and most important event, which means errors and destructive confirmations rather than branding and promotion** — a prioritisation that is frequently decided by whoever owns the page.
- **And test it by looking away and back — whatever you see first is what is actually emphasised. 'Look away and back' is a five-second test, and what you see first is what the design emphasises regardless of intent**, which makes it a fast and unarguable check.

4 Change blindness

PEOPLE RELIABLY FAIL TO NOTICE LARGE CHANGES OUTSIDE THEIR FOCUS OF ATTENTION

A substantial change to a scene, made during a saccade or a brief interruption, goes unnoticed by most observers — including changes to the very object they were nominally looking for, which is the finding that makes the effect so much stronger than intuition suggests.

So a status message appearing in a corner is not feedback — the user was looking at the button they just clicked, and the change happened somewhere else entirely — which is Module 01 §1's evaluation gap arriving as a perceptual mechanism rather than as a design oversight.

Which means feedback has to appear where the attention already is — at the point of interaction, on the control that was operated, at the cursor — **or it has to move, which overrides the blindness** because motion is preattentive (§3).

And it is why 'we showed a notification' is not a defence: showing is not the same as being perceived, and the difference is measurable — which is Module 01 §2's non-reading argument in a perceptual register.

- **At the point of interaction, or moving:** noticed reliably, which is the only combination you can depend on.
- **In the periphery, static, and not differing in a preattentive feature: reliably missed, however large it is —** size is not sufficient, which surprises people.
- **In a position where something always appears: missed, because the position has been habituated — which is why anything in a banner position is invisible,** and is a mistake worth avoiding when you put genuine content there.
- **And during a transition or a page change:** missed, **because the change blindness window is exactly then —** so a message that appears as part of a navigation is the worst available timing.
- **So place feedback at the cursor, at the field, or on the control that was operated — which is almost always technically possible and almost never done,** because a global notification area is easier to build. **Habituation to position is why anything in the advertising position is unseen,** including the genuine content you put there hoping it would be read.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Johnson — <i>Designing with the Mind in Mind</i> , chapters 1 through 5	The whole module , with the research cited and the effect sizes given. Library copy.
Ware — <i>Information Visualization: Perception for Design</i>	§2 and §3 in depth — the standard reference, and CSCE 679's primary text as well. Library copy.

Resource	What it covers for this module
Healey — Perception in Visualization (free)	§3's preattentive features, with interactive demonstrations — the conjunction search effect is worth experiencing rather than reading about.
Simons & Levin — Change blindness (free)	§4 in the original — and the magnitude of the effect is the part that changes how you design feedback.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Take a form you did not design** and identify where spacing contradicts the intended grouping.
- 2 **Fix it with spacing alone** and compare.
- 3 **Convert three interfaces to greyscale** and report what stops working.
- 4 **Find a colour-only encoding** in software you use.
- 5 **Build a pop-out search task** and measure your own time against the number of distractors.
- 6 **Build the conjunction version** and measure again.
- 7 **Count the emphasised elements** on three screens of your own software.
- 8 **Look away and back** from a screen you designed, and note what you saw first.
- 9 **Find a status message you have missed** and say where it should have appeared.
- 10 **Move one piece of feedback to the point of interaction** and test whether it is noticed.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name the six grouping principles and say which is strongest.
- 2 What happens when spacing contradicts intended grouping?
- 3 Why is a cluttered interface harder to use rather than merely uglier?
- 4 What is colour good and bad for, and how many categories?
- 5 Why must a design work in greyscale?
- 6 What are preattentive features, and what happens with a conjunction?
- 7 State the one-dimension rule and the countable defect.
- 8 What is the five-second emphasis test?
- 9 State change blindness and why a corner message is not feedback.

10 Give four cases and whether each will be noticed.